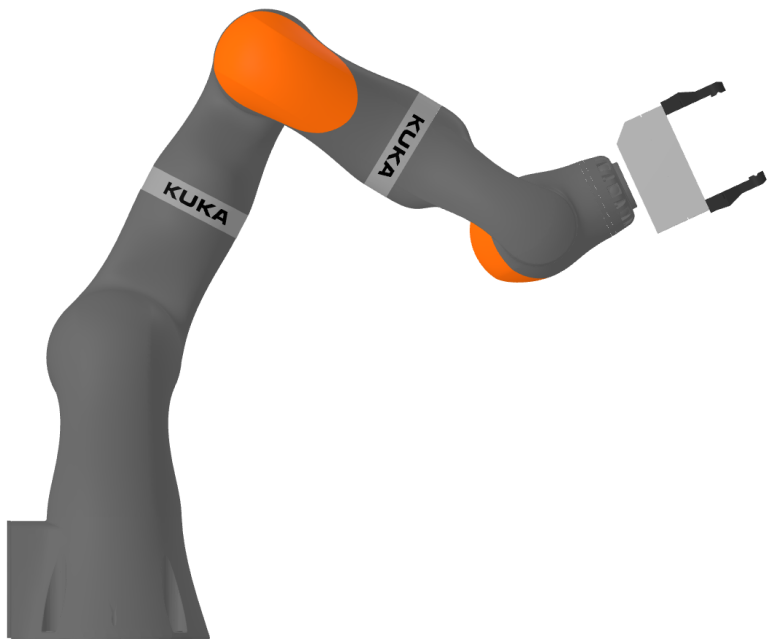
Manipulation in Clutter

Part 1: Contact



MIT 6.4210/2

Robotic Manipulation

Fall 2025, Lecture 7

Massachusetts Institute of Technology

Course Roadmap

1. Hardware basics (iiwa + WSG)
2. Basic pick and place
    - Kinematics / Jacobian-based control
    - Moved a single known object assuming known pose
3. Geometric perception
    - Pose estimation from depth for a single known object

This week:

- More complex scenes (many objects / diverse objects)
- Still relatively simple manipulation (pick and place)

"Clutter clearing" (an early project at TRI)



https://people.csail.mit.edu/russt/uploads/clutter_clearing.html

https://manipulation.csail.mit.edu/data/ falling_bricks_2d.html?height=400px

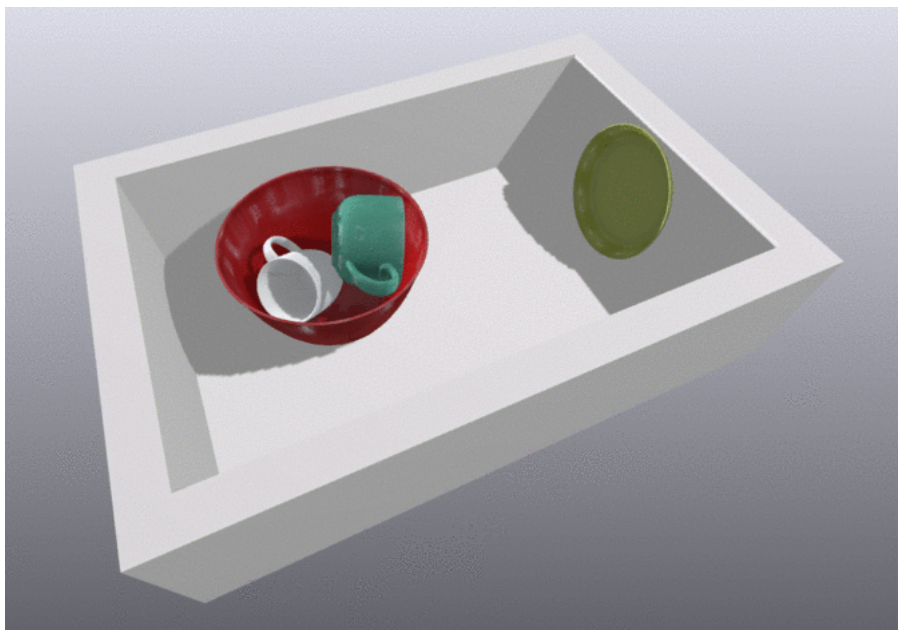


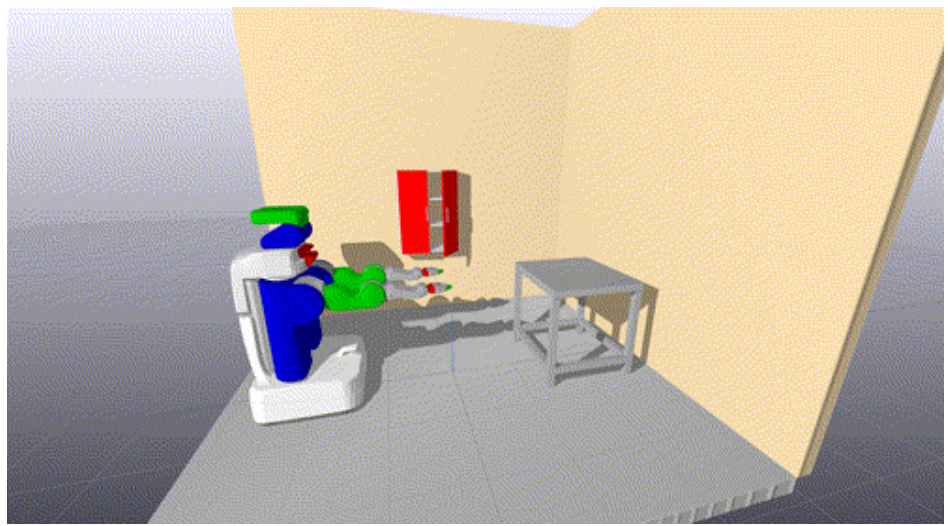
YCB Benchmarks – Object and Model Set




CHEEZ-IT

How do you generate random kitchens?





Probability distributions over environments. Work by Greg Izatt

- Spatial forces add when they are applied to the same body in the same frame, e.g.:

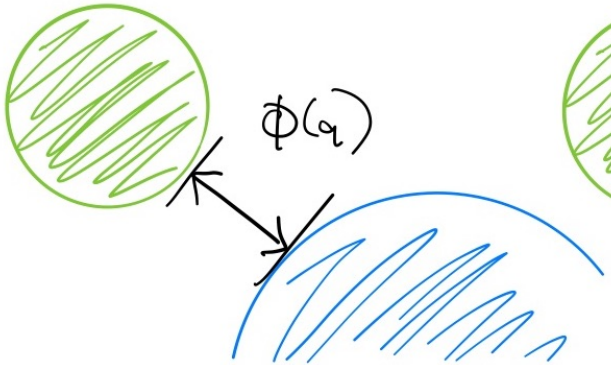
$$F^{B_p}_{\text{total},C} = \sum_i F^{B_p}_{i,C}.$$

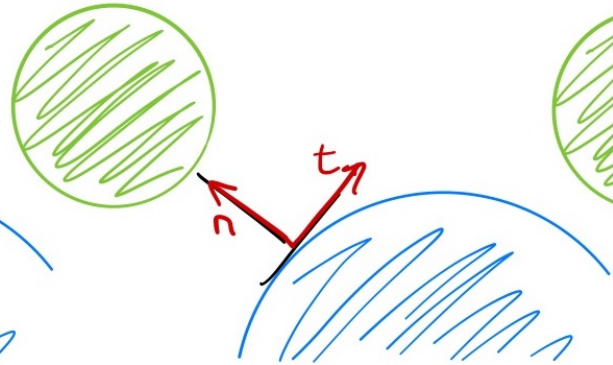
- Shifting a spatial force from one application point, B_p, to another point, B_q, uses the cross product:

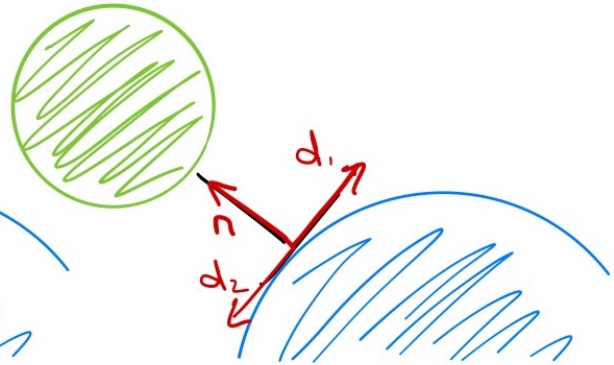
$$f_C^{B_q} = f_C^{B_p}, \qquad \tau_C^{B_q} = \tau_C^{B_p} + {}^{B_q}p_C^{B_p} \times f_C^{B_p}.$$

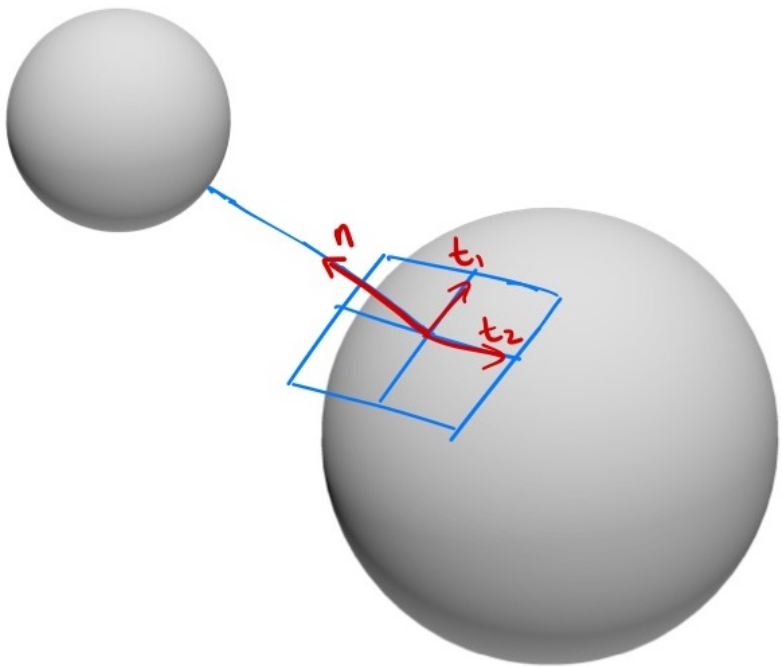
- As with all spatial vectors, rotations can be used to change between the "expressed-in" frames:

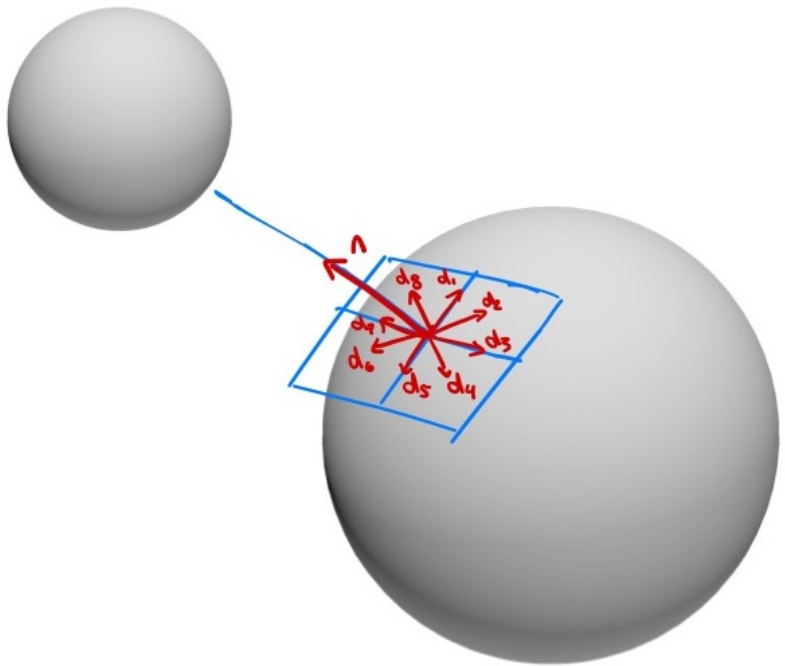
$$f_D^{B_p} = {}^{D}R^{C} f_C^{B_p}, \qquad \tau_D^{B_p} = {}^{D}R^{C} \tau_C^{B_p}.$$


$\phi(q)$


n
t_1

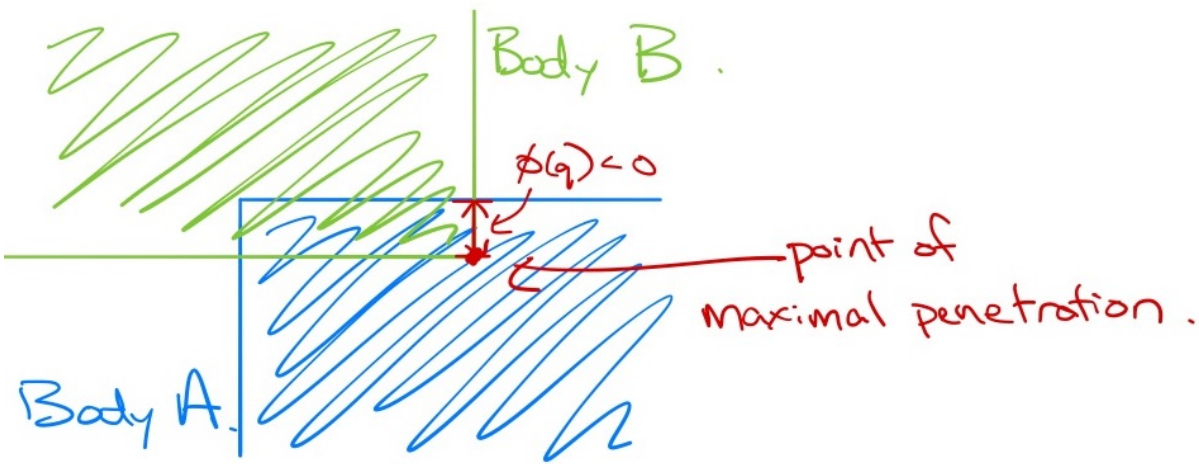

n
d_1
d_2


n
t_1
t_2


n
d_8
d_1
d_2
d_3
d_4
d_5
d_6
d_7
d_4

https://manipulation.csail.mit.edu/data/ falling_bricks_2d.html?height=400px

Rich collision geometries

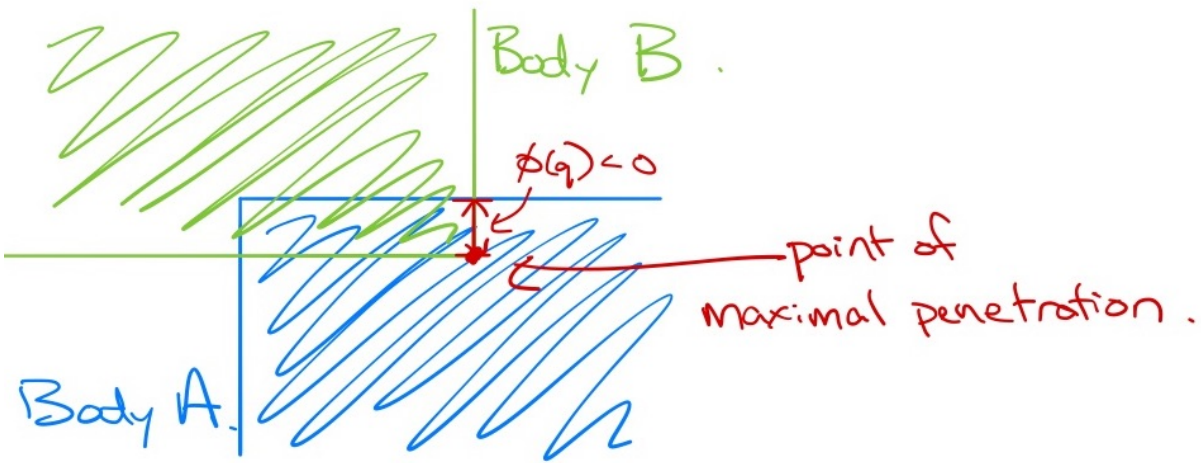


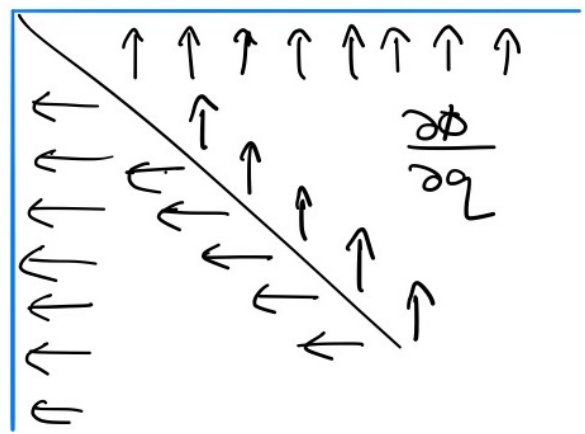
"Point contact" as implemented in Drake



Green arrow is the force on the red box due to the overlap with the blue box.

"Point contact" as implemented in Drake

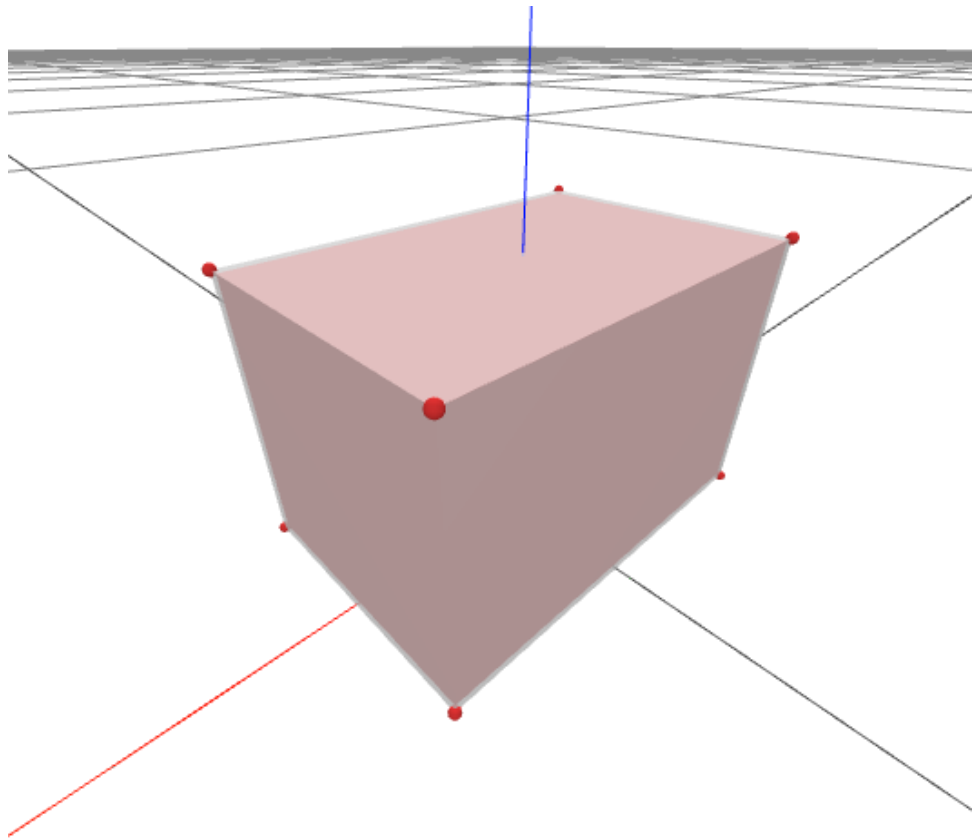




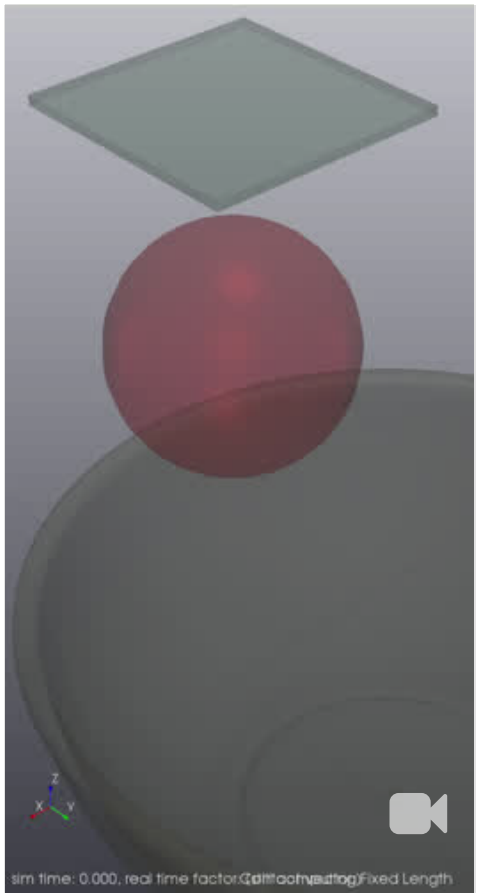
Multi-point contact

Many heuristics for using multiple points...

you can get more robust simulations with curated collision geometry



"Hydroelastic contact" as implemented in Drake

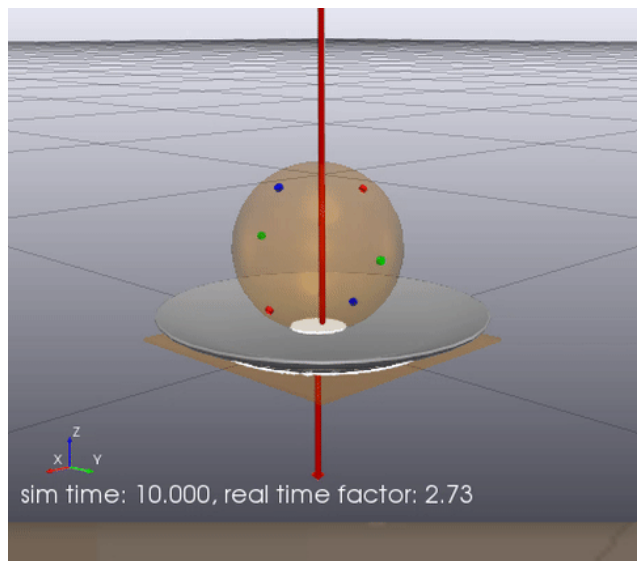


A pressure field model for fast, robust approximation of net contact force and moment between nominally rigid objects

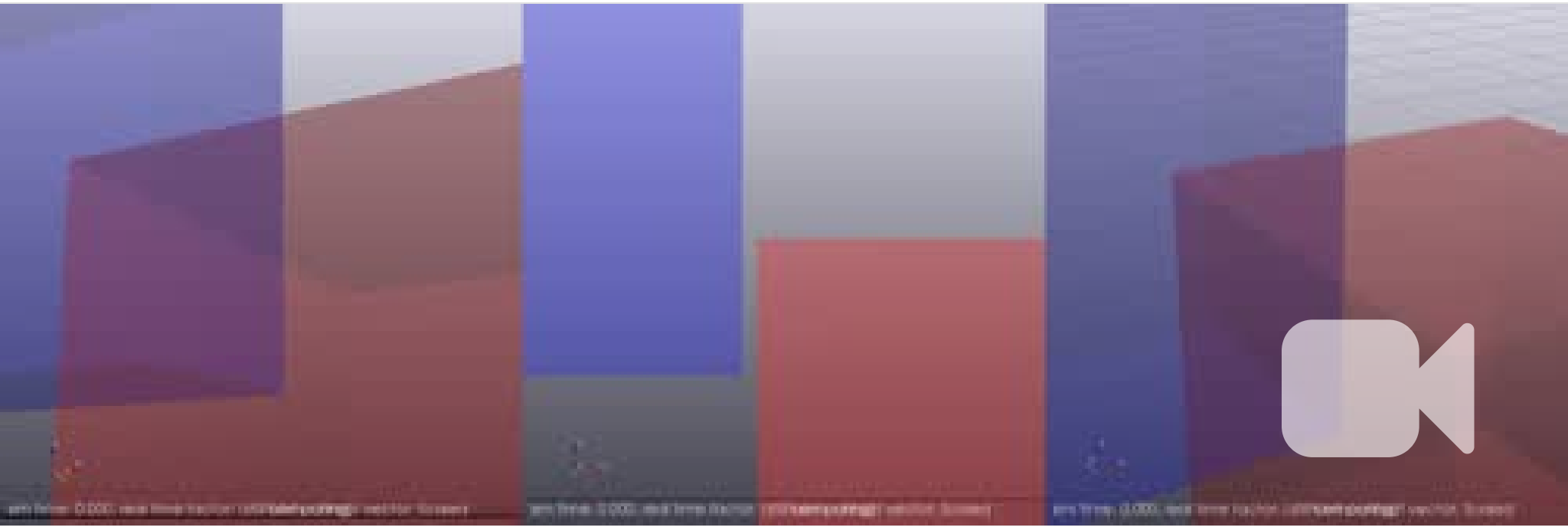
Ryan Elandt[1,2], Evan Drumwright[1], Michael Sherman[1], Andy Ruina[2]

TOYOTA RESEARCH INSTITUTE

major contributions from Damrong Guoy, Sean Curtis, Rick Cory, Alejandro Castro, ...



"Hydroelastic contact" as implemented in Drake



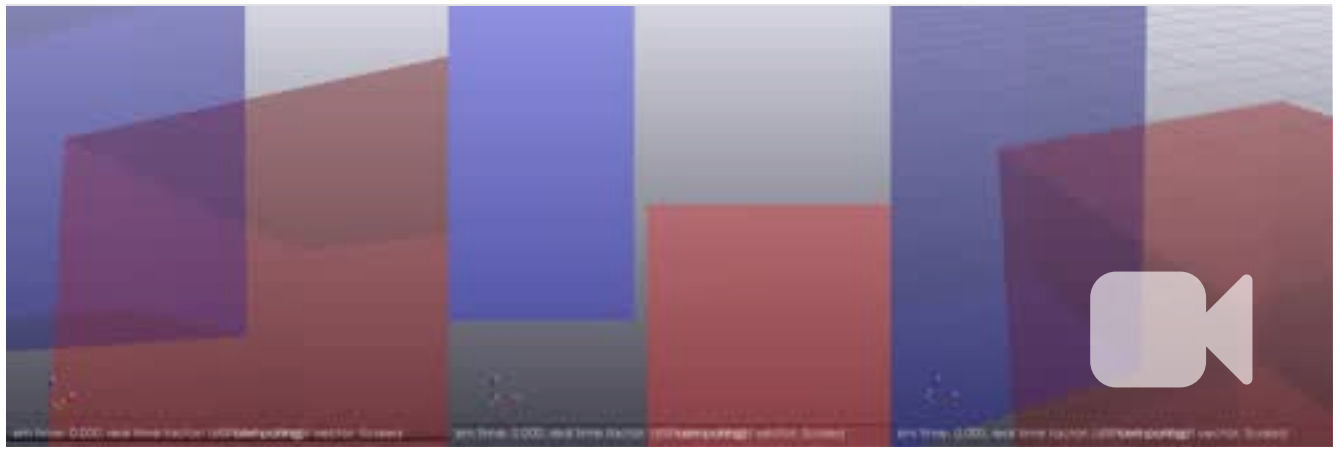
Both boxes are soft.

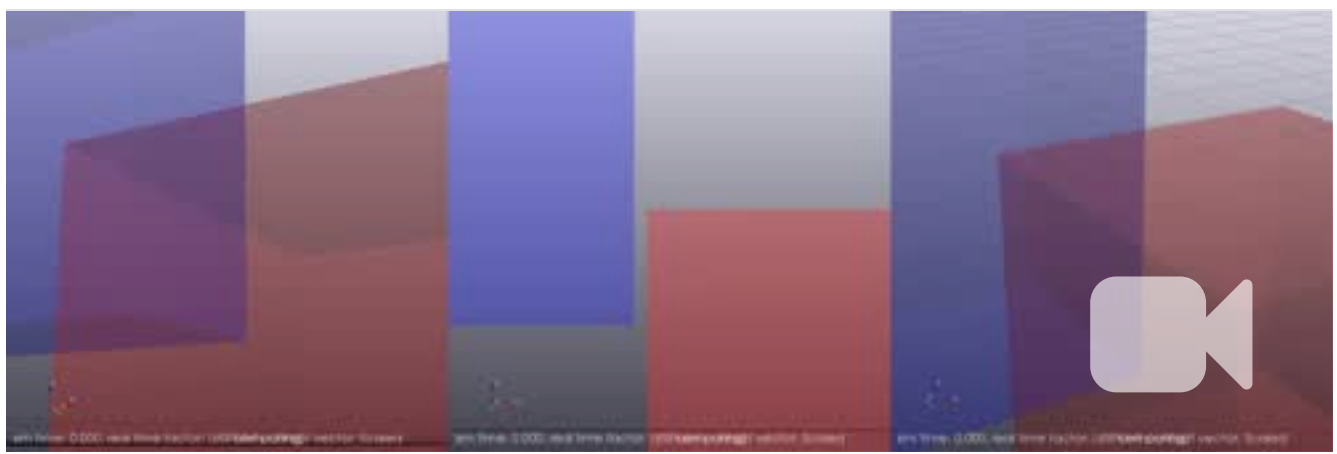
Point contact vs hydroelastic

Point contact
(discontinuous)

vs

Hydroelastic
(continuous)

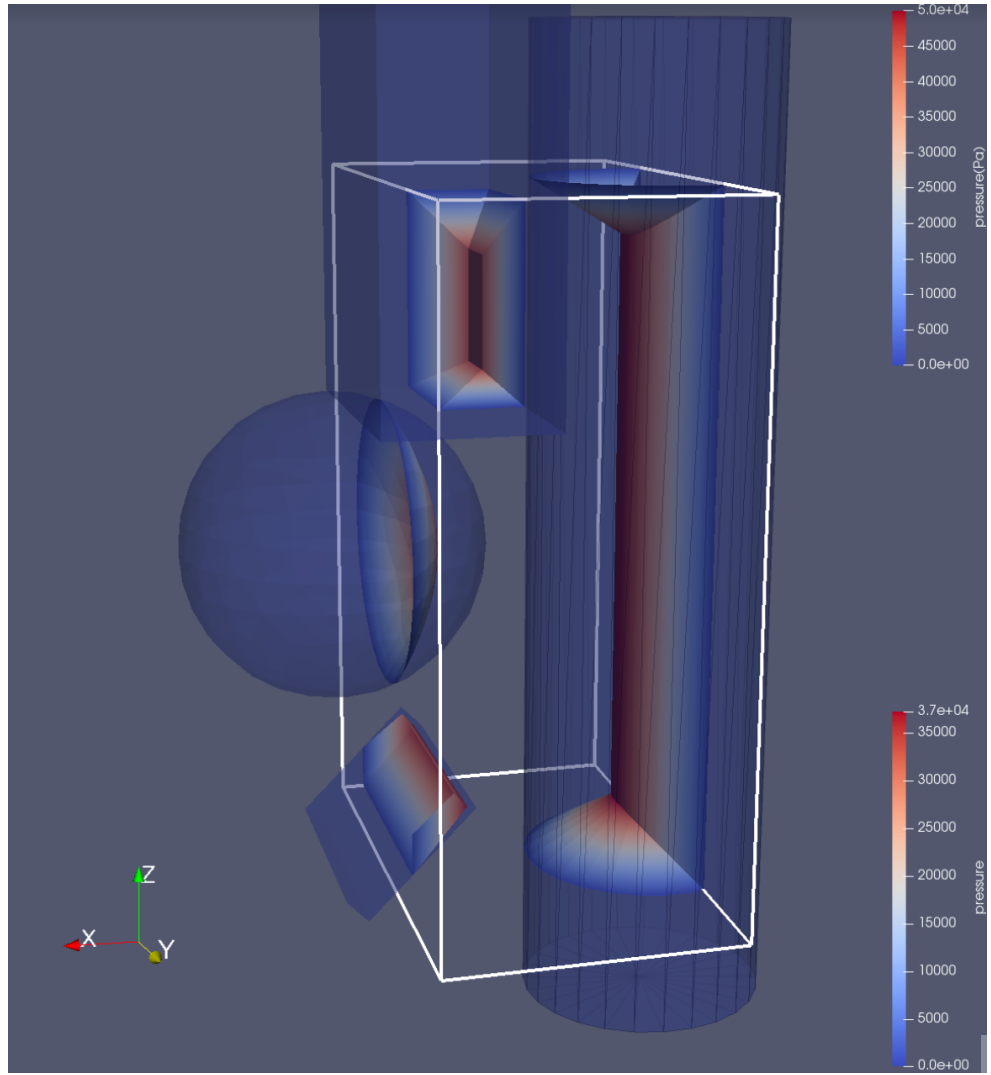




Hydroelastic is

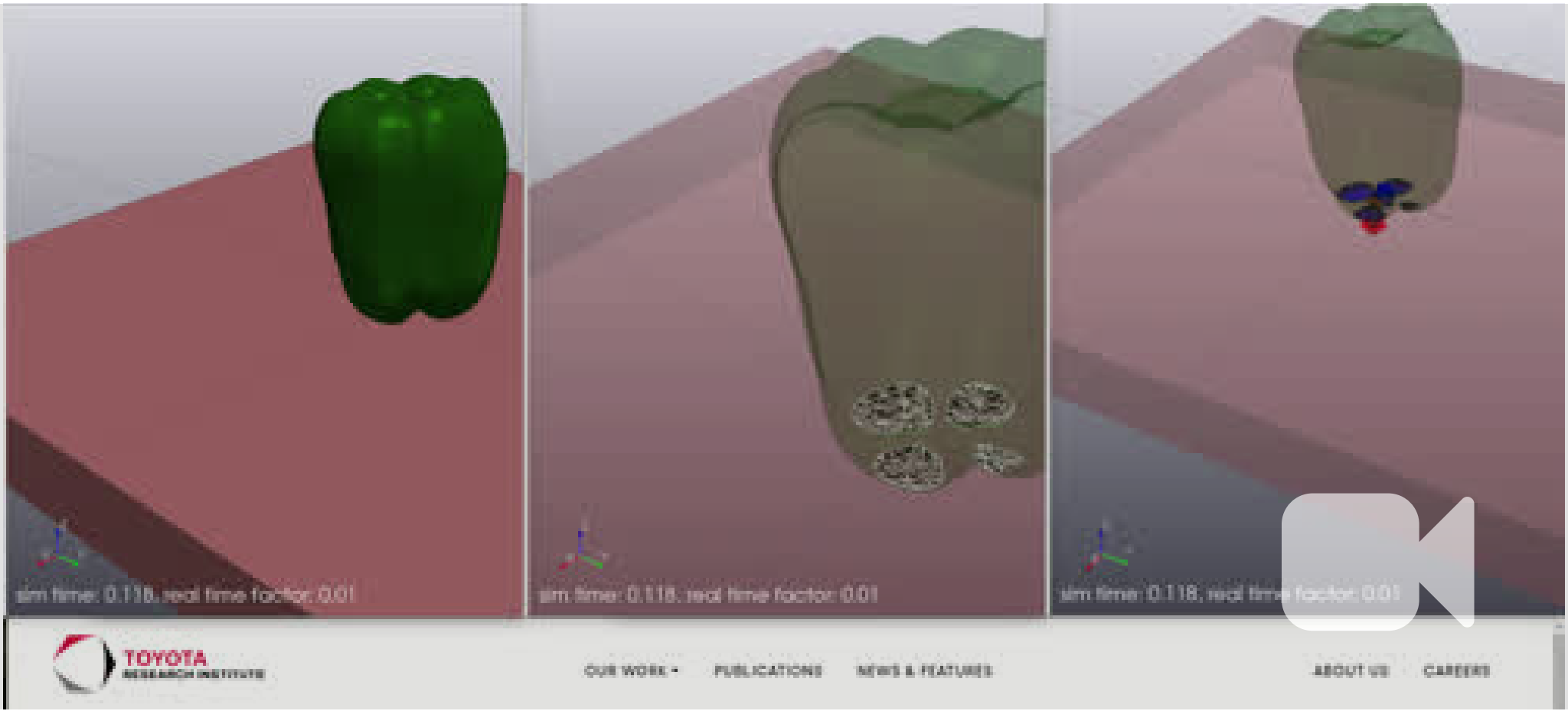
- more expensive than point contact
- (much) less expensive than finite-element models

State-space (for simulation, planning, control) is the original rigid-body state.



"Hydroelastic contact" as implemented in Drake




sim time: 0.118, real time factor: 0.01
sim time: 0.118, real time factor: 0.01
sim time: 0.118, real time factor: 0.01
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